

BAH Migration Runbook

traya-api-server → *traya-app-backend* · revised 3 September 2026 · v3, with measured production traffic

Build-A-Habit is **two systems sharing one write path**: the legacy 3/7/21 streak flow, which carries effectively 100% of production traffic, and the v85 Habit Tracker revamp, which is deployed but has almost no installed base. This document is the ground truth, the measured traffic, the difference between the two systems, the full endpoint inventory, and a ticket-by-ticket order to work through.

Source	server/components/BAH · 5,582 LOC across 12 modules
Endpoints	38, registered by buildAHabbitRoute(app) at server/app.js:283 — mounted on the app root, no /bah prefix
Deployments	TWO: traya-api-server-prod-prod (app) and traya-internal-api-service-prod-prod (CRM)
Traffic source	Last9 APM, span counts by http.route, 27 Aug – 3 Sep 2026
Client impact	None — public URLs unchanged at every step. No mobile release required.

00 : the earlier migration was never merged to master

A prior BAH migration effort exists, dated 25 June 2026, with a spec at docs/specs/2026-06-25-bah-endpoints-migration.md. It is **not on origin/master**. It exists only on origin/development (commits b9bbea75, a663c811), along with all the code it describes.

Service	origin/master (production)	origin/development
bahService.js	Present	Present
bahArchivedProductsService.js	Present	Present
kitTrackerPageService.js	Present	Present
habitTrackerStateResolver.js	Present	Present
scratchCardService.js	Present	Present

Service	origin/master (production)	origin/development
bahWriteService.js	ABSENT	Present
bahLogService.js	ABSENT	Present
bahActivityLogService.js	ABSENT	Present
bahStreakBalanceService.js	ABSENT	Present
bahMedicinesService.js	ABSENT	Present
bahValidator.js, bahExpiryService.js	ABSENT	Present

Production bahRoutes.js carries exactly four routes — /bah/coin/redeem, /bah/archive-product, /bah/unarchive-product, /bah/coin/habit-tracker-credit — plus scratchCardRoutes and the three /config/cms/kit-tracker-* endpoints. The five migrated BAH endpoints do not exist in production at all.

The June port is stale, not merely unmerged

server/components/BAH/ has taken **43 commits since 20 June 2026** — the entire v85 build, scratch cards, lifelines, archived products, the 15-day check-in nudge, the kit-tracker cache invalidation, and the appBackendService pooling refactor all landed after that port was written. Treat the development code as a **reference implementation, not a merge candidate**. Promoting it blind would roll back two and a half months of work.

BAH runs in two deployments

The CRM console does not call traya-api-server-prod-prod. It calls traya-internal-api-service-prod-prod, which serves the same BAH routes from the same component. Any route change lands in both. Measured over 7 days on the internal service:

Endpoint	7-day requests
GET /bahHistory/:caseId	75,312
GET /rewardBalance/:customerId	2,257
GET /streakMaster	365
POST /extraRewardsToUser/:caseId	5
PUT /syncRewardBalanceWithShopFlo/:caseId	3

01 — Measured traffic: what is actually

in use

Span counts from Last9, grouped by http.route, service traya-api-server-prod-prod, 7 days. If head sampling is enabled these understate absolute volume; relative ordering holds either way.

The ten endpoints that carry the product

Endpoint	7-day requests	Requests/day	rpm
GET /streakAndRewardBalance	1,162,104	166,015	115.3
GET /bahLogForGivenDate	771,511	110,216	76.5
GET /coinTransaction	476,051	68,007	47.2
GET /bah/:userId/calendar	402,045	57,435	39.9
GET /rewardBalance/:caseId	358,163	51,166	35.5
POST /activityLogForBAH	174,163	24,880	17.3
GET /medicinesForHowToUse	106,199	15,171	10.5
GET /latestOrderHowtoUseV2/:caseId	36,336	5,191	3.6
PUT /multipleActivityLogForBAH	25,184	3,598	2.5
GET /latestRoutineV2/:caseId	12,800	1,829	1.3

Near-dead — under 40 requests a day

Endpoint	7 days	30 days
GET /bahDetailDateWiseLikePagination	271	613
GET /latestMedicines	163	371
GET /kit-tracker-page	22	70
GET /activityLogsBAH	7	18
PUT /activityLogForBAH	4	—
GET /kit-tracker-calendar	1	2
GET /kit-tracker-badges	1	2

Zero requests in 7 days — 17 endpoints

Checked across both deployments.

Endpoint	30-day count	Note
GET /faqs	0	Static JSON
GET /termsAndConditions	0	Static JSON
GET /coinRewardHistory	1	Superseded by /coinTransaction
GET /newBahFlowEligibility	0	Returns a hardcoded true
GET /bahHistoryWeb/:caseId	0	
GET /bah-challenges-cx-data	0	Campaign appears to be over
GET /latestOrderHowtoUse/:caseId	0	v1; only V2 gets traffic
POST /addCoinBalance	0	Destructive tooling
GET /shopflo/txn/:caseId	0	CRM
POST /shopflo/debitReward	0	CRM
POST /shopflo/creditReward	0	CRM
POST /coinScript/:streakMasterId	0	CRM bulk tool
POST /uploadUsageDetails	0	Admin content-ops

Endpoint	30-day count	Note
POST /archiveProductForBAH	0	v85 feature, dormant
POST /unarchiveProductForBAH	0	v85 feature, dormant
GET /bah/scratch-card	1	v85 feature, dormant
POST /bah/scratch-card/:id/reveal	19	v85 feature, dormant

Six of these are v85 features with no installed base — dormant, not dead. The other eleven have no traffic in 30 days and no version explanation.

Net: 23 of 38 BAH endpoints run at under 40 requests a day. Ten endpoints carry essentially all the traffic.

02 — App version: which API is live for which version

Version distribution among BAH users

From /bahLogForGivenDate (771,036 requests carrying appVersion over 7 days; every BAH user hits this screen).

App version	Requests	Share
83	598,655	77.6%
81	154,654	20.1%
71–79	15,006	1.9%
82	2,705	0.35%
85 and above	21	0.003%
80, 84	0	—
70 and below	0	—

Fleet-wide (/homePageConfig, 895,254 requests) agrees: v80–84 is 98.3%, v71–79 is 1.7%, and **v85 and above is 58 requests, or 0.006%**.

The live app population is **v83 and v81**, with a 2% tail on v71–79. v85 exists as a real build on both platforms (iOS user agent Traya/5 CFNetwork, Android okhttp/4.12.0) but is not distributed.

Version-to-API matrix

Cohort	Share of fleet	Endpoints it calls
v85 and above	0.006%	/kit-tracker-page, /kit-tracker-calendar, /kit-tracker-badges, /bah/scratch-card, /bah/scratch-card/:id/reveal, /archiveProductForBAH, /unarchiveProductForBAH, and POST /activityLogForBAH with isHabitTracker true. Confirmed, not inferred: every one of the 22 kit-tracker-page calls carries appVersion=85, and no other version calls it.
v71 to v84	~100%	/bahLogForGivenDate, POST /activityLogForBAH (legacy branch), PUT /multipleActivityLogForBAH, /streakAndRewardBalance, /coinTransaction, /bah/:userId/calendar, /rewardBalance/:caseId, /medicinesForHowToUse, /latestOrderHowtoUseV2, /latestRoutineV2
v70 and below	0%	Extinct. No requests in 7 days.

Only /bahLogForGivenDate, /kit-tracker-page and /streakAndRewardBalance send a version parameter. The other endpoints carry none, so their version attribution is inferred from the fleet distribution plus the code gates rather than measured directly.

Three version gates are now dead branches

Gate	Location	Reality
showBahV3 = Number(appVersion) >= 71	/bahLogForGivenDate	Always true. The non-v3 else path is unreachable.
Number(version) >= 71 for bannerWidgetData and the streak-broken recompute	getStreakAndRewardBalance	Always true. The bannerWidgetData null path is unreachable.

Gate	Location	Reality
appVersion >= 85 archived-product filtering	getBahLogForGivenDate	Taken 21 times a week out of 771,000.

03 — The v85 revamp: two reward economies, one shared write endpoint

The v85 Habit Tracker is not a redesign of BAH. It is a **different reward economy running on the same collections**, owned by app-backend, deployed to production, and carrying almost no traffic.

Dimension	Legacy BAH (v71–84, api-server, ~100% of traffic)	v85 Habit Tracker (app-backend, 0.006% of fleet)
Read surface	/streakAndRewardBalance, /bahLogForGivenDate, /bahDetailDateWiseLikePagin ation, /bah/:userId/calendar	/config/cms/kit-tracker-page, -calendar, -badges, /bah/scratch-card
Reward ladder	Milestones at day 3 / 7 / 21, paying 100 / 400 / 2000 coins. O8+ cohort gets 200 / 600 / 2500. Nothing on non-milestone days.	EVERY DAY PAYS. Daily tier by streak week: 30 / 40 / 50 / 60 / 70 coins. Ladder days 7 / 14 / 21 / 28 / 35 replace the daily coin with 100 / 150 / 200 / 250 / 300.
Streak cap	Resets to 1 on reaching day 21.	Ladder ends at day 35; earning pauses after 35 valid logs in a kit window.
Missing a day	Gap of more than 2 days breaks the streak outright.	3 LIFELINES PER KIT auto-bridge missed days via a BulIMQ delayed job. The streak only breaks once the budget is spent.
Coin delivery	Credited at log time, into reward_transactions and the Shopflo wallet.	Reward days MINT AN UNSCRATCHED SCRATCH CARD; coins credit on reveal. 7-day scratch window, then forfeited.
Coin expiry	Order-service driven: coin/expiry/month/:userId.	Fixed 90 days.

Dimension	Legacy BAH (v71–84, api-server, ~100% of traffic)	v85 Habit Tracker (app-backend, 0.006% of fleet)
Kit coupling	Soft: a 45-day medicine window and "Log and Earn is locked" copy.	Hard: kit windows of 30 days per kit plus 10-day buffer, reorder banner from day 30n minus 9, pause at 30n plus 5.
Idempotency	NONE. Re-running the reward path re-credits.	Enforced: credit_remarks is habit-tracker-daily-DATE and is checked before crediting.
Cohort and extras	O8+ experiment, male and female streak-restart bonuses, 15-day check-in nudge, community share CTA at 7/21.	None of those apply. Instead: 21 per-kit badges at 30 logs each, and a 13-state page resolver.

The seam

Both economies are selected inside **one api-server endpoint**. POST /activityLogForBAH reads isHabitTracker straight from the request body; shouldUseHabitTrackerCredit() then either calls app-backend's scratch-card mint (v85) or runs the local 3/7/21 credit (legacy). GET /bahLogForGivenDate branches on appVersion >= 85, and both app versions call it for the dose list.

So api-server owns the write path and the dose list for v85 users too. The revamp moved the display surface to app-backend and left the writes behind. **Finishing the migration is about moving that write path, not about rebuilding v85.**

Correction to the earlier draft

An earlier version of this document said v85 was "already live and already served by app-backend" and listed /bah/:userId/calendar, /bahDetailDateWiseLikePagination, /activityLogsBAH and /coinRewardHistory as endpoints superseded by v85 and therefore sunset candidates. The traffic data contradicts that. v85 is deployed but not rolled out, so **nothing is being superseded**: /bah/:userId/calendar (402,045 a week) and /coinTransaction (476,051 a week) are the third and fourth busiest BAH endpoints and must be migrated, not retired.

04 — Endpoint inventory: all 38 routes

A · Already proxying in production

Endpoint	7d	Serves	Logic	Target
POST /archiveProductForBAH	0	v85	Removes a product from the logging list; last-product rule enforced downstream.	POST /bah/archive-product
POST /unarchiveProductForBAH	0	v85	Re-adds a removed product. Logging history untouched.	POST /bah/unarchive-product
GET /bah/scratch-card	0	v85	Scratch-card dashboard. JWT userId forwarded; app-backend 404/410 pass through.	GET /bah/scratch-card
POST /bah/scratch-card/:id/reveal	0	v85	Scratches a card and credits coins. Where v85 rewards actually land.	POST /bah/scratch-card/:id/reveal
GET /kit-tracker-page	22	v85	The v85 home surface: 13-state resolver, day strip, reward screen, reorder banner, feedback card.	GET /config/cms/kit-tracker-page
GET /kit-tracker-calendar	1	v85	All months first-log to today, grouped as months[]. Redis cached, 5 min.	GET /config/cms/kit-tracker-calendar
GET /kit-tracker-badges	1	v85	21 per-kit badges, earned at 30 valid logs per kit window.	GET /config/cms/kit-tracker-badges

B · Ported on development, absent from production

Endpoint	7d	Serves	Logic	Reference port
GET /streakAndRewardBalance	1,162,104	v71–84	The heaviest read: six parallel queries for balance, streak log, reward history, BAH eligibility, non-void orders and user/case. Derives the O8+ and streak-restart cohorts, the banner widget, the full modal set, the community share CTA at 7/21, and the 15-day check-in nudge (female, 1 order, treatment week 3, W1/W3 call not completed, reading Postgres user_order_reminders).	GET /bah/streak-reward-balance, bahStreakBalanceService (dev only)
GET /bahLogForGive nDate	771,511	All	The dose-logging screen for one date. Loads the day's log plus getLatestMedicines, decides isPostApiNeeded vs isPutApiNeeded, splits products into daily vs weekly dosage with per-slot copy from	GET /bah/log-for-date, bahLogService (dev only)

Endpoint	7d	Serves	Logic	Reference port
			dosageCode, filters archived products at appVersion 85+, appends coin-expiry text, streak state and the challenge banner.	
POST /activityLogForBAH	174,163	All	The one write both economies share. Validates, drops archived products, resolves the IST check-in date, writes user_activity_logs_for_bah, busts the kit-tracker Redis key. Branches on isHabitTracker: V85 runs synchronously and mints a scratch card; LEGACY emits an in-process event, credits 3/7/21 milestones, resolves the streak-restart cohort, mirrors to Shopflo. Both mark the build_a_habit tasks and enqueue the lifeline check.	POST /bah/activity-log, bahWriteService (dev only)
PUT /multipleActivityLogForBAH	25,184	All	Flips morning/evening check-ins for 4 to	PUT /bah/multiple-activity-log

Endpoint	7d	Serves	Logic	Reference port
			5 products in one bulkWrite. Deliberately forgiving: a missing log, an empty payload and a yesterday conflict all return success.	
PUT /activityLogForBAH	4	All	Single-product version of the above. Effectively dead — four requests in seven days.	PUT /bah/activity-log, bahActivityLogService (dev only)

C · Not started

Endpoint	7d	Logic	Call
GET /coinTransaction	476,051	Paginated ledger (6 per page) that also synthesises Expired rows for credits past expire_at with coins unused, joined to the streak display name.	MIGRATE. Sorts and slices the full history in memory; convert to an aggregation.
GET /bah/:userId/calendar	402,045	Calendar or streak mode. Range starts at the first delivered order or first_date_of_log; paints each day inactive / missed / logged / coin / active, grouped by month.	MIGRATE. Takes userId from the PATH, not the token — a live IDOR at 57,000 requests a day. Fix first.
GET /rewardBalance/:caseId	358,163	Aggregate of unexpired, unspent credits minus debits.	MIGRATE, but keep a local copy — customer/handler imports it.
GET /medicinesForHowTo	106,199	How-to-use list from the latest	MIGRATE. medicinesForHowToU

Endpoint	7d	Logic	Call
Use		non-void/unknown orders, 50-day or bulk-duration window, Last Delivered / Expired labels.	seService already exists on master.
GET /latestOrderHowtoUseV2/:caseId	36,336	Proxies recommendation-service how-to-use/:caseId, then attaches reminder info from the newest customer_activityLog with reminder_days.	MIGRATE. Already thin.
GET /latestRoutineV2/:caseId	12,800	Pass-through to recommendation-service routine/:caseId.	MIGRATE. Smallest live endpoint — good first cutover.
GET /bahDetailDateWiseLikePagination	271	Day-by-day timeline, minus 90 to plus 21 days, with a rolling 1-to-21 serial and Unlock/Missed labels.	DELETE. 39 requests a day.
GET /latestMedicines	163	The 45-day medicine window. Still hot in-process (bahLogForGivenDate, user_task) but almost never called as a route.	DELETE THE ROUTE, keep the function.
GET /activityLogsBAH	7	Month grid of logs with IST log times.	DELETE. One request a day.
GET /coinRewardHistory	0	Merged credit and debit ledger. Reads its filters from req.body on a GET.	DELETE. Superseded by /coinTransaction.
GET /faqs, GET /termsAndConditions	0	Static JSON files.	DELETE.
GET /newBahFlowEligibilit	0	Hardcoded true; real check commented	DELETE.

Endpoint	7d	Logic	Call
y		out.	
GET /bahHistoryWeb/:caseId	0	Web rewards view via caseId.	DELETE.
GET /bah-challenges-cx-data	0	7-day challenge card; rank from Redis BAH!CHALLENGE!RANK keys.	DELETE after confirming the campaign is over.
GET /latestOrderHowtoUse/:caseId	0	v1 how-to-use with the 10-branch topical combination table and sku_usage aggregation.	DELETE. Only V2 gets traffic, so the sku_usage model is never needed in app-backend.

D · CRM and admin — stays in api-server

Endpoint	7d	Logic	Call
GET /bahHistory/:caseId	75,312	CRM agent view: coin ledger, streak balance, month log grid, latest medicines, Shopflo balance, and whether this agent may grant coins.	KEEP. Fans out to four BAH reads; will call app-backend for some after Wave 3.
GET /rewardBalance/:customerId	2,257	Balance lookup on the internal service.	KEEP.
GET /streakMaster	365	Lists active is_for_superuser streak masters for the CRM bonus dropdown.	KEEP.
POST /extraRewardsToUser/:caseId	5	Manual coin grant; custom amount only for existing_coins_streak . Writes the txn and mirrors to Shopflo.	KEEP.
PUT	3	Reconciles internal	KEEP.

Endpoint	7d	Logic	Call
/syncRewardBalance WithShopFlo/:caseId		balance against the Shopflo wallet.	
POST /addCoinBalance	0	Deletes all reward transactions for a user, then rebuilds from CMS.	DELETE after CRM sign-off. Destructive.
GET /shopflo/txn/:caseId	0	Wallet read by phone number.	DELETE after CRM sign-off.
POST /shopflo/debitReward	0	Manual wallet debit.	DELETE after CRM sign-off.
POST /shopflo/creditReward	0	Manual wallet credit.	DELETE after CRM sign-off.
POST /coinScript/:streakMasterId	0	Bulk grant from a CSV of case ids.	DELETE after CRM sign-off.
POST /uploadUsageDetails	0	Upserts one sku_usage document by variantId.	DELETE after content-ops sign-off.

05 — The tether: functions other components import directly

BAH is not self-contained. handler.js cannot be deleted unless these stay reachable in-process.

Function	Imported by	Disposition
calculateDaysDifference	attendance, user_task/handler, user_task/userTask, customer/handler	Move to server/utils. Pure date helper.
getBahRunningLogDay	user_task/handler, user_task/userTask	Keep local; one small streak_logs read.
getLatestMedicines	user_task/handler	Keep local even after the route is deleted.
getUserRewardBalance	customer/handler	Keep local; cheap aggregation.

Function	Imported by	Disposition
saveRewardTransaction	customer/handler	Keep local. A SECOND WRITER of reward_transactions.

Lifelines exist twice: api-server has lifelineQueue / lifelineWorker / lifelineDecision / lifelineState / applyLifeline / lifelineReconcileCron, and app-backend has the original plus habit_tracker_lifeline_logs on master. The api-server BullMQ worker is commented out at server/setup/bullmq.js lines 8 to 9, but enqueueLifelineForLog is still called from the live write path, so the api-server copy retires only when POST /activityLogForBAH does.

06 — The plan: 22 tickets, picked up one at a time

Every ticket is independently shippable and independently revertible. Rules that apply to all of them: app-backend deploys to master first; one ticket per deploy; a write never shares a deploy with a read; and every route change lands in both deployments.

The ordering principle is **shrink, then secure, then migrate in ascending blast radius**. Waves 0 and 1 remove 17 endpoints and fix two live defects before any traffic moves, which cuts the surface from 38 endpoints to 21 and makes every later ticket smaller.

Wave 0 — Shrink the surface (no migration, pure deletion)

#	Ticket	Scope	Verify
1	Delete 7 dead app and web endpoints	/faqs, /termsAndConditions, /coinRewardHistory, /newBahFlowEligibility, /bahHistoryWeb/:caseId, /bah-challenges-cx-data, /latestOrderHowtoUse/:caseId (v1). All zero over 30 days. Remove the routes, their handlers, medicineTrackerFAQs.json and	Last9 shows no 404s appearing on those paths for 7 days after.

#	Ticket	Scope	Verify
		medicineTrackerTCs.json.	
2	Retire 6 CRM and admin endpoints	/addCoinBalance, /shopflo/txn, /shopflo/debitReward, /shopflo/creditReward, /coinScript/:streakMasterId, /uploadUsageDetails. Zero over 30 days on both deployments. NEEDS CRM and content-ops sign-off first — low-frequency tools can be quarterly.	Written confirmation from CRM ops, then the same 7-day 404 watch.
3	Delete 4 near-dead endpoints after a deprecation notice	PUT /activityLogForBAH (4 a week), /activityLogsBAH (7), /bahDetailDateWiseLikePagination (271), and the GET /latestMedicines route (163 — keep the function, it is used in-process). Announce, wait one app release cycle, then remove.	Traffic still under threshold at removal time.
4	Remove the dead version branches	The sub-v71 fallback paths in getStreakAndRewardBalance and getBahLogForGivenDate. Zero traffic below v71, so both are unreachable. Deleting them shrinks the two largest functions before they are ported.	Unit tests still pass; no behaviour change for v71+.

Wave 0 exit: 17 endpoints gone. The migration surface is 21 endpoints, of which 10 carry real

traffic.

Wave 1 — Fix what is broken, before moving anything

#	Ticket	Scope	Verify
5	Fix the IDOR on /bah/:userId/calendar	The route takes userId from the path, not the token, at 57,435 requests a day. Bind it to req.userId and ignore the path parameter, or reject a mismatch. Independent of the migration and should not wait for it.	A token for user A cannot read user B's calendar. Traffic unchanged.
6	Make the legacy credit path idempotent	v85 guards on credit_remarks = habit-tracker-daily-DATE. The legacy 3/7/21 path has no guard. Add an equivalent key and a uniqueness index on reward_transactions before any write moves. This is the precondition for ticket 19.	Replaying a log in staging produces exactly one credit.
7	Decide the CCD question	api-server emits CCD_UPDATE for ACTIVITY_LOG and COIN_CREDITED. app-backend has no equivalent. Get a written decision from the data owners: drop them, keep the emit on the api-server proxy after a successful response, or give app-backend an event hop. No code, but it blocks	A decision recorded on the ticket.

#	Ticket	Scope	Verify
		ticket 19.	
8	Re-baseline the development port	Diff each of the seven dev-only services against its api-server counterpart as it stands today. Expect gaps in archived-product filtering, the scratch-card mint branch, lifeline enqueue, kit-tracker cache invalidation, the 15-day nudge, the pooling refactor, and the two August getStreakAndReward Balance simplifications. Decide per service: rebase or rewrite. Default to rewrite.	A gap list and a rebase-or-rewrite decision per service.
9	Build the parity harness	Two cohorts. LEGACY: new user, mid-streak, day 21, streak-broken, O8+ eligible, streak-restart male and female, BAH-locked, 15-day-check-in eligible. V85: kit-arriving intro, day 1, day 6, day 7 unscratched, day 7 claimed, lifeline used, all lifelines used, earning paused, new kit delivered. Script every endpoint against both, diff field by field including nulls and array order. Extend test/experiments/bah	One command produces a full diff report.

#	Ticket	Scope	Verify
		ServiceParity.test.js.	

Wave 2 — Migrate reads, ascending blast radius

Each ticket is one endpoint. Ship it, watch a full day, then take the next. The order is by traffic, smallest first, so the first cutover risks 1,829 requests a day and the last risks 166,015.

#	Endpoint	Req/day	Notes for this ticket
10	GET /latestRoutineV2/:caseId	1,829	A pass-through to recommendation-service. The rehearsal ticket: proves the proxy shape, the env vars and the rollback drill on the smallest live endpoint. Needs RECOMMENDATION_SERVICE_BASE_URL and V2_FORM_DATA_TOKEN in app-backend.
11	GET /latestOrderHowtoUseV2/:caseId	5,191	Same proxy plus a customer_activityLog read for reminder_days. Note that deleting the v1 route in ticket 1 means the sku_usage model is never needed in app-backend.
12	GET /medicinesForHowToUse	15,171	medicinesForHowToUseService already exists on master. Mostly exposure plus a parity diff.
13	GET /rewardBalance/:customerId	51,166	Cheap aggregation. Keep the local getUserRewardBalance for customer/handler. The internal-service /rewardBalance/:cust

#	Endpoint	Req/day	Notes for this ticket
			omerId is a separate route and stays.
14	GET /bah/:userId/calendar	57,435	Only after ticket 5. Forward the token identity, never the path parameter.
15	GET /coinTransaction	68,007	Convert the in-memory sort and slice to an aggregation while porting. Include the synthesised Expired rows in the parity diff.
16	GET /bahLogForGivenDate	110,216	The largest read, and it drags getLatestMedicines with it. Land bahLogService and bahMedicinesService on master, reconciled per ticket 8. Forward appVersion so the 85+ archived-product branch keeps working. getLatestMedicines stays in api-server for user_task.
17	GET /streakAndRewardBalance	166,015	The heaviest endpoint and the widest surface: cohorts, modals, banner, community CTA, 15-day nudge. Needs COMMUNITY_BASE_URL and the user_order_reminders Postgres model in app-backend. CRM /bahHistory depends on it, so check

#	Endpoint	Req/day	Notes for this ticket
			/bahHistory after this lands.

Wave 3 — Migrate the writes

#	Ticket	Req/day	Notes
18	PUT /multipleActivityLogForBAH	3,598	An idempotent field update on an existing document — the safest write and the rehearsal for the next ticket. Keep the forgiving semantics: a missing log, an empty payload and a yesterday conflict all answer success. Turning those into errors is a client-visible regression.
19	POST /activityLogForBAH	24,880	Its own deploy, alone. Preconditions: tickets 6, 7, 8, 9 all closed. Preserve the isHabitTracker branch verbatim — v85 synchronous so the scratch card is in the response, legacy fire-and-forget. Move the kit-tracker-calendar cache invalidation with the write. Move enqueueLifelineForLog and retire the api-server lifeline modules in the same change. Verify exactly one credit per log per cohort in staging before production.

Wave 4 — Close out

#	Ticket	Scope
20	Re-point CRM /bahHistory	It fans out to getRewardCoinHistory, getStreakAndRewardBalance , getActivityLogs and getLatestMedicines. After Wave 2 some of those live in app-backend; make the internal service call them there rather than keeping a second copy.
21	Extract the tether and delete the dead half	Move calculateDaysDifference to server/utils. Keep getBahRunningLogDay, getUserRewardBalance, saveRewardTransaction and getLatestMedicines in a small internal module. Delete the now-unreachable write path and helpers, the duplicated lifeline modules and their tests, and the commented BullMQ registrations. Trim the BAH test suite to what remains.
22	Decide the v85 rollout	Not migration work. v85 is code-complete on both platforms and carries 0.006% of traffic. Either roll it out, or set a date to remove the dormant surface. Until one of those happens the codebase carries two reward economies, and every ticket above has to keep both alive.

07 — Risks

Severity	Risk	What happens	Mitigation
Blocking · security	Live IDOR on /bah/:userId/calendar	userId is read from the path rather than the token, on an endpoint serving 57,435 requests a day. Any authenticated user can read any other user's logging calendar.	Ticket 5, ahead of all migration work.
Blocking · data integrity	Legacy crediting is not idempotent	The legacy 3/7/21 path has no guard, and both services can write reward_transactions and call the Shopflo credit API. A retry or a half-deployed fleet double-credits.	Ticket 6 before ticket 19. Flip the write endpoint alone and verify one credit per log per cohort.
Blocking · downstream data	CCD events stop firing	ACTIVITY_LOG and COIN_CREDITED are emitted from api-server only. app-backend has no equivalent and the June port omitted them, so customer-computed-data goes quietly stale from ticket 19 onward.	Ticket 7 — a recorded decision from the data owners.
Blocking · correctness	Promoting the stale port	Merging development's BAH services to master would overwrite two and a half months of v85 work with a June snapshot.	Ticket 8. Never promote a dev-branch BAH service without a line-by-line diff.
Medium · correctness	The economy branch is client-supplied	isHabitTracker arrives in the request body and selects which reward economy	Forward it explicitly, and consider deriving it from appVersion instead so the

Severity	Risk	What happens	Mitigation
		runs. After ticket 19 that choice crosses a service boundary.	economy follows the client build.
Medium · coupling	Shared Redis keys	saveActivityLogs busts the kit-tracker-calendar key that app-backend owns. bahChallengeData reads BAH!CHALLENGE!RANK keys written elsewhere.	The invalidation moves with the write in ticket 19. Grep both repos for each key before deleting.
Medium · contract	Error shapes diverge	api-server answers with an err object and a 500 for almost everything; app-backend answers with a message and a real status, forwarded through the proxy.	Check what the app does with each response before flipping. Preserve the old shape where a client parses err.
Medium · scope	Two deployments	The CRM runs the same BAH component in traya-internal-api-service. A route flip in api-server lands there too.	Every ticket's verification step checks both services in Last9, not just api-server.
Low · scope	v85 dormancy	Every ticket has to keep two reward economies working for a feature used by 0.006% of the fleet.	Ticket 22 — force a roll-out-or-remove decision.

08 — Environment and gaps

app-backend

Variable	Needed by
RECOMMENDATION_SERVICE_BASE_URL	Ticket 10
V2_FORM_DATA_TOKEN	Ticket 10
S3_IMAGE_BASE_URL	Ticket 16 (product and banner assets)
COMMUNITY_BASE_URL	Ticket 17 — net-new, the 7/21-day share CTA
ORDER_SERVICE_BASE_URL	Ticket 19 (legacy coin expiry)
SHOPFLO_WALLET_API_ENPOINT	Ticket 19
SHOPFLO_WALLET_ISSUER_ID	Ticket 19
SHOPFLO_WALLET_MERCHANT_ID	Ticket 19
SHOPFLO_WALLET_API_KEY	Ticket 19

api-server, proxy side

Variable	Note
APP_BACKEND_SERVER_BASE_URL	Required in both deployments
APP_BACKEND_MAX_SOCKETS	Pool size, defaults to 25
HTTP_DEFAULT_TIMEOUT_MS	Client default

Models still missing in app-backend

Missing	Needed for
user_order_reminders (Postgres)	The 15-day check-in nudge inside ticket 17
sku_usage (Mongo)	NO LONGER NEEDED once ticket 1 deletes /latestOrderHowtoUse v1
multer, csvtojson	NO LONGER NEEDED once ticket 2 deletes /coinScript

09 — Per-cutover checklist

Run this for every ticket in Waves 2 and 3, not once for the migration.

1. **Diff both cohorts** — legacy and v85 — api-server local vs app-backend, field by field including nulls and array ordering.
2. **Deploy app-backend to master first.** The proxy target must exist in production before

the proxy does.

3. **Confirm the env vars** that ticket needs are present in the environment you just deployed to.
4. **Flip the route** and ship. One ticket per deploy; a write never shares a deploy with a read.
5. **Watch a full day in Last9** — request count, error rate and p95 on that route, in BOTH traya-api-server-prod-prod and traya-internal-api-service-prod-prod. For writes, also the count of reward transactions per log, split by cohort.
6. **Only then delete** the local handler, and only once nothing in section 05 imports it.